CS 631: DATA MANAGEMENT SYSTEMS DESIGN

EXERCISES

SQL
Data changes, Views and Complex Constraints in SQL.

<u>EXERCISE 1. (Data Changes)</u> (*Fundamentals of Database Systems*, Elmasri, Navathe, Addison-Wesley). Specify the following data change operations on the COMPANY database using the data change commands of SQL.

1. Insert <‘Robert’, ‘F’, ‘Scott’, ‘943775543, ‘1952-06-21’, ‘2365 Newcastle Rd, Bellaire, TX’, M, 58000, ‘888665555’, 1> into EMPLOYEE.
2. Insert <‘ProductA’, 4, ‘Belaire’, 2> into PROJECT.
3. Insert <‘Production’, 4, ‘943775543’, ‘1998-10-01’> into DEPARTMENT.
4. Insert <‘677678989’, NULL, ’40.0’> into WORKS_ON.
5. Delete the WORKS_ON tuples with ESSN = ‘333445555’
6. Delete the EMPLOYEE tuple with SSN = ‘987654321’
7. Modify the MGRSSN and MGRSTARTDATE attribute values of the DEPARTMENT tuple with DNUMBER = 5 to ‘123456789’ and ‘1999-10-01’ respectively.
8. Modify the SUPERSSN of the EMPLOYEE tuple with SSN = ‘999887777’ to ‘943775543’.
9. Modify the HOURS attribute value of the WORKS_ON tuple with ESSN = ‘999887777’ and PNO = 10 to ‘5.0’.

<u>EXERCISE 2.</u>

Write SQL update statements to do the following on the schema of the following database.

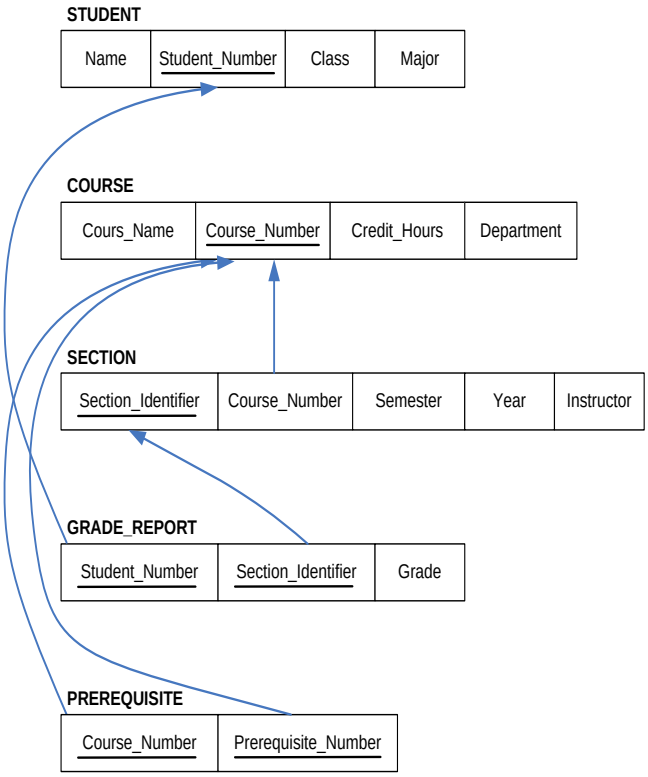
(a) Insert a new student <'Johnson', 25, 1, 'MATH'> in the database.

(b) Change the class of student 'Smith' to 2.

(c) Insert a new course <'Knowledge Engineering','CS4390', 3,'CS'>.

(d) Delete the record for the student whose name is 'Smith' and student number is 17.

The Relational Schema of the University Database



STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

EXERCISE 3. (Views) (*Fundamentals of Database System*, Elmasri, Navathe, Addison-Wesley)

Specify the following views in SQL on the COMPANY database schema.

1. A view that has the department name, manager name, and manager salary for every department.
2. A view that has the employee name, supervisor name, and employee salary for each employee who works in the 'Research' department.
3. A view that has the project name, controlling department name, number of employees, and total hours worked per week on the project for each project.
4. A view that has the project name, controlling department name, number of employees, and total hours worked per week on the project for each project *with more than one employee working on it*.

EXERCISE 4. (Complex Constraints) (From the book of Ramakrishnan and Gehrke)

Consider the following database schema:

STUDENTS (SNUM: *integer*, SNAME : *string*, MAJOR : *string*, LEVEL : *string*, AGE : *integer*)
CLASS (NAME : *string*, MEETS_AT : *time*, ROOM : *string*, FID : *integer*)
ENROLLED (SNUM : *integer*, CNAME : *string*)
FACULTY (FID : *integer*, FNAME : *string*, DEPTID : *integer*)

The meaning of these relations is straightforward; for example, ENROLLED has one record per student-class pair such that the student is enrolled in the class.

1. Write the SQL statements required to create these relations, including appropriate versions of all primary and foreign key i
2. Express each of the following integrity constraints in SQL unless it is implied by the primary and foreign key constraint; If the constraint cannot be expressed in SQL, say so. For each constraint, state what operations (inserts, deletes, and updates on specific relations) must be monitored to enforce the constraints.

   (a) Every enrolled class has a minimum enrollment of 5 students and a maximum enrollment of 30 students.

   (b) In each room, at least one class meets.

   (c) Every faculty member must teach at least two courses.

   (d) Only faculty in the department with DEPTID = 33 teach more than three courses

   (e) Every student must be enrolled in the course called 'Math 101'.

   (f) Two classes cannot meet in the same room at the same time.

   (g) No department can have more than 10 faculty members.

   (h) The number of CS majors must be more than the number of math majors.